

Stat 517
Homework # 2

1. Let F have an F distribution with parameters r_1 and r_2 . Argue that $1/F$ has an F distribution with parameters r_2 and r_1 .
2. Let $\bar{X}$ be the mean of a random sample of size n from a distribution that is $N(\mu, 9)$. Find n such that $P(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.90$, approximately
3. Let two independent random samples, each of size 10, from two normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ yield $\bar{x} = 4.8$, $s_1^2 = 8.64$, $\bar{y} = 5.6$, $s_2^2 = 7.88$. Find a 95% confidence interval for $\mu_1 - \mu_2$.
4. Let $X_1, X_2, \dots, X_n, X_{n+1}$ be a random sample of size $n + 1$, $n > 1$, from a distribution that is $N(\mu, \sigma^2)$. Let $\bar{X} = \sum_{i=1}^n X_i/n$ and $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2/(n - 1)$. Find the constant c so that the statistic $c(\bar{X} - X_{n+1})/S$ has the t-distribution. If $n = 8$, determine k such that $P(\bar{X} - kS < X_9 < \bar{X} + kS) = 0.80$. The observed interval $(\bar{x} - ks, \bar{x} + ks)$ is often called an 80% *prediction interval* for X_9 .
5. Let X have a pdf of the form $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, zero elsewhere, where $\theta \in \{\theta : \theta = 1, 2\}$. To test the simple hypothesis $H_0 : \theta = 1$ against the alternative simple hypothesis $H_1 : \theta = 2$, use a random sample X_1, X_2 of size $n = 2$ and define the critical region to be $C = \left\{ (x_1, x_2) : \frac{3}{4} \leq x_1 x_2 \right\}$. Find the power function of the test.
6. Let X have a binomial distribution with the number of trials $n = 10$ and with p either $1/4$ or $1/2$. The simple hypothesis $H_0 : p = 1/2$ is rejected, and the alternative simple hypothesis $H_1 : p = 1/4$ is accepted, if the observed value of X_1 , a random sample of size 1, is less than or equal to 3/ Find the significance level and the power of the test.
7. Let X_1, X_2 be a random sample of size $n = 2$ from the distribution having pdf $f(x; \theta) = (1/\theta) e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. We reject $H_0 : \theta = 2$ and accept $H_1 : \theta = 1$ if the observed values of X_1, X_2 , say x_1, x_2 , are such that

$$\frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} \leq \frac{1}{2}.$$

Here $\Omega = \{\theta : \theta = 1, 2\}$. Find the significance level of the test and the power of the test when H_0 is false.

8. Assume that the weight of cereal in a “10-ounce box” is $N(\mu, \sigma^2)$. To test $H_0 : \mu = 10.1$ against $H_1 : \mu > 10.1$, we take a random sample of size $n = 16$ and observe that $\bar{x} = 10.4$ and $s = 0.4$.
- (a) Do we accept or reject H_0 at the 5% significant level?
 - (b) What is the approximate p-value of this test?

1. Let F have an F distribution with parameters r_1 and r_2 . Argue that $1/F$ has an F distribution with parameters r_2 and r_1 .

$$F = \frac{U/r_1}{V/r_2} \text{ where } \begin{matrix} U \sim \chi^2(r_1) \\ V \sim \chi^2(r_2) \end{matrix} \text{ and } U \text{ and } V \text{ are independent.}$$

$$\frac{1}{F} = \frac{V/r_2}{U/r_1} \xrightarrow{\substack{\text{chi-square var.} \\ \text{with } r_2 \text{ d.o.f.}}} \frac{1}{F} \sim F(r_2, r_1)$$

If $F \sim F(r_1, r_2)$, then $1/F \sim F(r_2, r_1)$

Normal Distribution

2. Let $\bar{X}$ be the mean of a random sample of size n from a distribution that is $N(\mu, 9)$. Find n such that $P(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.90$, approximately

$$N(\mu, \sigma^2) \rightarrow N(\mu, 9)$$

$$\text{variance} = 9 = \sigma^2$$

$$SD = 3 = \sigma$$

$$\text{sample mean } \bar{X} : \sim N\left(\mu, \frac{\sigma^2}{n}\right) \rightarrow N\left(\mu, \frac{9}{n}\right)$$

$$\text{standard error or SD } SE(\bar{X}) : \frac{\sigma}{\sqrt{n}} \rightarrow \frac{3}{\sqrt{n}}$$

$\bar{X}$ is centered at μ ,
so $\bar{X} - \mu = 0$ with $SD = 3/\sqrt{n}$

$$P(\bar{X} - 1 < \mu < \bar{X} + 1)$$

$$P(-1 < \mu - \bar{X} < 1)$$

$$P(1 > \bar{X} - \mu > -1)$$



To standardize, divide all 3 by $SD = 3/\sqrt{n}$ $P(-1 < \bar{X} - \mu < 1) = 0.90$

$$P\left(\frac{-1}{3/\sqrt{n}} < \frac{\bar{X} - \mu}{3/\sqrt{n}} < \frac{1}{3/\sqrt{n}}\right) = 0.90$$

For 90% CL: critical Z-value is 1.645.

$$\frac{1}{3/\sqrt{n}} = \frac{\sqrt{n}}{3} = 1.645$$

$$n = 24.35 \approx 25$$

3. Let two independent random samples, each of size 10, from two normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ yield $\bar{x} = 4.8$, $s_1^2 = 8.64$, $\bar{y} = 5.6$, $s_2^2 = 7.88$. Find a 95% confidence interval for $\mu_1 - \mu_2$.

Two pop. have the same σ^2 . we can use 'pooled variance':

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{(10-1)(8.64) + (10-1)(7.88)}{10+10-2} = \frac{148.68}{18}$$

$$s_p^2 = 8.26$$

$$t_{\alpha/2, \text{dof}} = t_{0.025, 18}$$

For 95% CL: critical t-value is 2.101.

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{8.26} \left(\sqrt{\frac{1}{10} + \frac{1}{10}} \right)$$

$$SE = 1.285$$

$$95\% \text{ CL for } (\mu_1 - \mu_2) = (\bar{x} - \bar{y}) \pm t_{0.025, 18} (SE)$$

$$= (4.8 - 5.6) \pm 2.101 (1.285)$$

$$= -0.8 \pm 2.70$$

$$95\% \text{ CL for } (\mu_1 - \mu_2) = (-3.50, 1.90)$$

4. Let $X_1, X_2, \dots, X_n, X_{n+1}$ be a random sample of size $n+1$, $n > 1$, from a distribution that is $N(\mu, \sigma^2)$. Let $\bar{X} = \sum_{i=1}^n X_i / n$ and $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)$. Find the constant c so that the statistic $c(\bar{X} - X_{n+1})/S$ has the t-distribution. If $n = 8$, determine k such that $P(\bar{X} - kS < X_9 < \bar{X} + kS) = 0.80$. The observed interval $(\bar{x} - ks, \bar{x} + ks)$ is often called an 80% prediction interval for X_9 .

$$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$$

$$X_{n+1} \sim N(\mu, \sigma^2)$$

$$a) \quad t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \rightarrow \frac{\bar{X} - X_{n+1}}{S\sqrt{(n+1)/n}} \sim t_{n-1}$$

$$\frac{c(\bar{X} - X_{n+1})}{S} = \sqrt{\frac{n}{n+1}} \left(\frac{\bar{X} - X_{n+1}}{S} \right) \sim t_{n-1}$$

$$c = \sqrt{\frac{n}{n+1}}$$

$$\bullet \text{ mean: } \mu - \mu = 0$$

$$\bullet \text{ variance: } \frac{\sigma^2}{n} + \sigma^2 = \sigma^2 \left(\frac{1}{n} + 1 \right) = \sigma^2 \left(\frac{n+1}{n} \right)$$

$$\bullet \text{ SD: } \sqrt{\sigma^2 \left(\frac{n+1}{n} \right)} = \sigma \sqrt{\frac{n+1}{n}}$$

$$= \sigma \sqrt{\frac{n+1}{n}}$$

$$b) \quad n=8, c = \sqrt{8/9} \rightarrow t_{8-1} = t_7$$

For 80% CL: critical t-value is 1.415.

$$\sqrt{8/9} (k) = 1.415$$

$$k \approx 1.50$$

standardize:

$$P(\bar{X} - kS < X_9 < \bar{X} + kS)$$

$$P(-kS < X_9 - \bar{X} < kS)$$

$$P(|\bar{X} - X_9| < kS) = 0.80$$

$$P\left(\frac{|\bar{X} - X_9|}{S} < k\right) = 0.80$$

$$P\left(\sqrt{\frac{8}{9}} \left(\frac{\bar{X} - X_9}{S}\right) < \sqrt{\frac{8}{9}} (k)\right)$$

5. Let X have a pdf of the form $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, zero elsewhere, where $\theta \in \{\theta : \theta = 1, 2\}$. To test the simple hypothesis $H_0 : \theta = 1$ against the alternative simple hypothesis $H_1 : \theta = 2$, use a random sample X_1, X_2 of size $n = 2$ and define the critical region to be $C = \left\{ (x_1, x_2) : \frac{3}{4} \leq x_1 x_2 \right\}$. Find the power function of the test.

All pairs (x_1, x_2) such that $x_1 x_2 \geq 3/4$ $\gamma(\theta) = P_\theta(\text{reject } H_0)$

$r(\theta) = P_\theta(x_1 x_2 \geq 3/4)$

Fixing x_1 and x_2 both work. we can fix x_1 : $x_1 x_2 \geq 3/4$

Since $0 < x < 1$: $x_2 \geq \frac{3}{4}x_1$ and $x_2 < 1$

Restriction 1 Bottom $\frac{3}{4}x_1 < x_2 < 1$

Restriction 2 Top $\frac{3}{4} < x_1 < 1$

x_1 x_2

$$r(\theta) = \int_{3/4}^1 \int_{3/4(4x_1)}^1 \theta^2 x_1^{\theta-1} \cdot x_2^{\theta-1} dx_2 dx_1$$

$f(x; \theta) = \theta x^{\theta-1}$, the joint pdf is the product:

$$f(x_1, x_2; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) = \theta x_1^{\theta-1} \cdot \theta x_2^{\theta-1}$$

$$f(x_1, x_2; \theta) = \theta^2 x_1^{\theta-1} \cdot x_2^{\theta-1}$$

$\theta=1$:

$$r(1) = \int_{3/4}^1 \int_{3/4(4x_1)}^1 1 dx_2 dx_1 = \int_{3/4}^1 \left(1 - \frac{3}{4x_1}\right) dx_1$$

$$= \left[x_1 - \frac{3}{4} \ln x_1 \right]_{3/4}^1 = 1 - \left(\frac{3}{4} - \frac{3}{4} \ln \frac{3}{4} \right)$$

$r(1) = 0.0342$

$\theta=2$:

$$r(2) = \int_{3/4}^1 \int_{3/4(4x_1)}^1 2^2 x_1^{2-1} x_2^{2-1} dx_2 dx_1$$

$$= \int_{3/4}^1 \int_{3/4(4x_1)}^1 4 x_1 x_2 dx_2 dx_1$$

$$= \int_{3/4}^1 4 x_1 \left[\frac{x_2^2}{2} \right]_{3/4(4x_1)}^1 dx_1 = \int_{3/4}^1 2 x_1 \left(1 - \frac{9}{16 x_1^2} \right) dx_1$$

$$= \left[x_1^2 - \frac{9}{8} \ln x_1 \right]_{3/4}^1 = 1 - \left(\frac{9}{16} - \frac{9}{8} \ln \frac{3}{4} \right)$$

$r(2) = 0.1139$

$$r(\theta) = \begin{cases} \frac{1}{4} + \frac{3}{4} \ln \frac{3}{4} & \theta=1 \\ \frac{7}{16} + \frac{9}{8} \ln \frac{3}{4} & \theta=2 \end{cases}$$

6. Let X have a binomial distribution with the number of trials $n = 10$ and with p either $1/4$ or $1/2$. The simple hypothesis $H_0 : p = 1/2$ is rejected, and the alternative simple hypothesis $H_1 : p = 1/4$ is accepted, if the observed value of X_1 , a random sample of size 1, is less than or equal to 3/ Find the significance level and the power of the test. Reject H_0 if $X \leq 3$
 $X=0,1,2,3$

$$H_0: p = 1/2$$

$$H_A: p = 1/4$$

a)

b)

$$a) \quad \alpha = P(\text{Reject } H_0 | H_0 \text{ true})$$

$$\begin{aligned} \alpha &= P(X \leq 3 | p = 1/2) = P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= \binom{10}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10} + \binom{10}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^9 + \binom{10}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^8 + \binom{10}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^7 \end{aligned}$$

$$\alpha = 0.1719$$

$$b) \quad \gamma = P(\text{Reject } H_0 | H_A \text{ true})$$

$$\begin{aligned} \gamma &= P(X \leq 3 | p = 1/4) = P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= \binom{10}{0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{10} + \binom{10}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^9 + \binom{10}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^8 + \binom{10}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^7 \end{aligned}$$

$$\gamma = 0.7759$$

7. Let X_1, X_2 be a random sample of size $n = 2$ from the distribution having pdf $f(x; \theta) = (1/\theta) e^{-x/\theta}, 0 < x < \infty$, zero elsewhere. We reject $H_0 : \theta = 2$ and accept $H_1 : \theta = 1$ if the observed values of X_1, X_2 , say x_1, x_2 , are such that

$$\frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} \leq \frac{1}{2}.$$

$$P(Y \leq y) = 1 - e^{-y/\theta} \left(1 + \frac{y}{\theta}\right)$$

Here $\Omega = \{\theta : \theta = 1, 2\}$. Find the significance level of the test and the power of the test when H_0 is false.

a) $\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$

b) $\gamma = P(\text{Reject } H_0 | H_A \text{ true})$

$$H_0: \theta = 2 \longrightarrow f(x; 2) = \frac{1}{2} (e^{-x/2})$$

$$H_A: \theta = 1 \longrightarrow f(x; 1) = e^{-x}$$

$$\begin{aligned} \text{Reject } H_0 &\longrightarrow \frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} \leq \frac{1}{2} \longrightarrow \frac{\frac{1}{2} (e^{-x_1/2}) \frac{1}{2} (e^{-x_2/2})}{e^{-x_1} e^{-x_2}} = \frac{1}{4} e^{-\frac{x_1}{2} - (-x_1) - \frac{x_2}{2} - (-x_2)} \\ &= \frac{1}{4} e^{x_1/2 + x_2/2} = \frac{1}{4} e^{\frac{1}{2}(x_1 + x_2)} \leq \frac{1}{2} \\ \frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} &= e^{\frac{1}{2}(x_1 + x_2)} \leq 2 \quad \text{Reject } H_0 \\ &\quad \times \ln \quad \times \ln \quad \times 4 \quad \times 4 \\ &= \frac{1}{2} (x_1 + x_2) \leq \ln 2 = \underline{x_1 + x_2 \leq 2 \ln 2} \end{aligned}$$

a) $\alpha = P(\text{Reject } H_0 | H_0 \text{ true})$
 $\downarrow \quad \downarrow$
 $\theta = 2 \quad y = 2 \ln 2$

$$P(Y \leq y) = 1 - e^{-y/\theta} \left(1 + \frac{y}{\theta}\right) = 1 - e^{-2 \ln 2 / 2} \left(1 + \frac{2 \ln 2}{2}\right) = 1 - e^{-\ln 2} (1 + \ln 2) = 1 - \frac{1}{2} (1 + \ln 2)$$

$\alpha = 0.1534$

b) $\gamma = P(\text{Reject } H_0 | H_A \text{ true})$
 $\downarrow \quad \downarrow$
 $\theta = 1 \quad y = 2 \ln 2$

$$P(Y \leq y) = 1 - e^{-y/\theta} \left(1 + \frac{y}{\theta}\right) = 1 - e^{-2 \ln 2} \left(1 + \frac{2 \ln 2}{1}\right) = \frac{3 - 2 \ln 2}{4}$$

$\gamma = 0.4034$

8. Assume that the weight of cereal in a "10-ounce box" is $N(\mu, \sigma^2)$. To test $H_0 : \mu = 10.1$ against $H_1 : \mu > 10.1$, we take a random sample of size $n = 16$ and observe that $\bar{x} = 10.4$ and $s = 0.4$. *one sample t-test*

(a) Do we accept or reject H_0 at the 5% significant level?

(b) What is the approximate p-value of this test?

$$\begin{aligned} H_0: \mu &= 10.1 & n &= 16 \text{ dof} = 15 \\ H_A: \mu &> 10.1 & \bar{x} &= 10.4 \\ & & s &= 0.4 \end{aligned}$$

$$\begin{aligned} a) \quad T &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \rightarrow T = \frac{10.4 - 10.1}{0.4/\sqrt{16}} = \frac{0.3}{0.4/4} \\ & & T &= 3 \end{aligned}$$

$$\begin{aligned} T &> t: \text{Reject} \\ \text{Right-tailed } \alpha &= 0.05: \\ t_{0.05, 15} &= 1.753 \\ 3 &> 1.753 \end{aligned}$$

Reject H_0 , evidence mean cereal weight > 10.1 oz

b) Calculator: 0.0045

$$P(T_{15} \geq 3) \approx 0.0045$$